



KINCOPPAL – ROSE BAY
SCHOOL OF THE SACRED HEART

2010

HIGHER SCHOOL CERTIFICATE
HALF YEARLY EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total marks – 72

- Attempt Questions 1 – 6
- All questions are of equal value

Question 1 (12 marks) Use a SEPARATE writing booklet **Marks**

(a) Solve for x , $\frac{3}{2-3x} \leq \frac{2}{3}$ **3**

(b) Solve the equation $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$ **3**

(c) Evaluate $\lim_{\theta \rightarrow 0} \frac{\tan \frac{\theta}{3}}{\theta}$ **2**

(d) If $f(x) = 2x^2 + x$, use the definition **2**

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

to find the derivative of $f(x)$ at the point where $x = a$

(e) Find the coordinates of the point P which divides the interval AB externally in the ratio 4:1, if A is $(1, 4)$ and B is $(3, -6)$ **2**

End of Question 1

Question 2 (12 marks)	Use a SEPARATE writing booklet	Marks
(a)	Use the substitution $u = 1 + x^2$ to evaluate $\int_1^{\sqrt{3}} 6x\sqrt{1+x^2} \, dx$	3
(b)	Let $f(x) = \ln(\tan x)$, $0 < x < \frac{\pi}{2}$ Show that $f'(x) = 2\operatorname{cosec}2x$	3
(c)	(i) Express $\cos 3t - \sqrt{3}\sin 3t$ in the form $R \cos(3t + \alpha)$ for some $R > \alpha$ and $0 < \alpha < \frac{\pi}{2}$	2
	(ii) Hence state the period and amplitude of $\cos 3t - \sqrt{3}\sin 3t$.	2
(d)	The word EQUATIONS contains all five vowels. How many 7-letter 'words' consisting of all five vowels can be formed from the letters of EQUATIONS?	2

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet **Marks**

- (a) Prove by mathematical induction, for $n \geq 2$, that **3**

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}.$$

- (b) Find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 3x \, dx$ **3**

- (c) The polynomial $p(x) = x^3 + ax^2 + bx + 12$ has a zero at $x = -1$ and has remainder 8 when divided by $x + 2$. Find the constants a and b . **3**

- (d) If α, β and γ are the roots of $x^3 - 6x^2 - 2x + 4 = 0$, find the values of:

(i) $\alpha + \beta + \gamma$ **1**

(ii) $\alpha^2 + \beta^2 + \gamma^2$ **2**

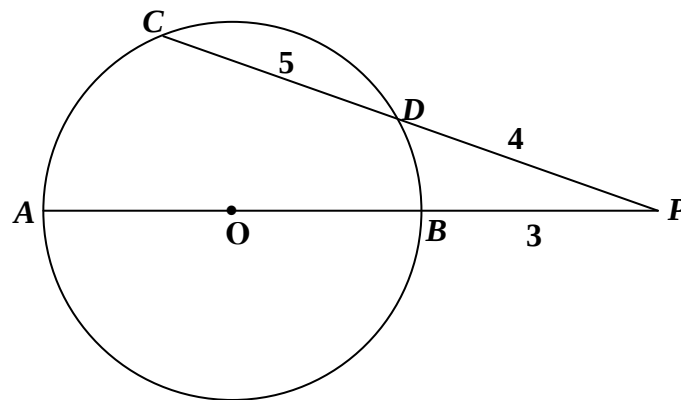
End of Question 3

Question 4 (12 marks)

Use a SEPARATE writing booklet

Marks

(a)

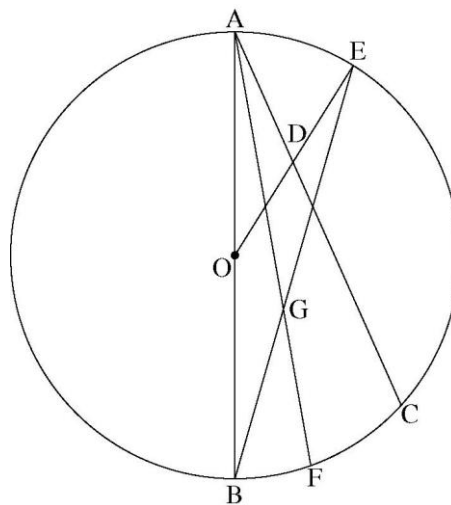


2

AB is the diameter of a circle, centre O .
 AB produced meets the secant CD at P .
 $CD = 5$, $DP = 4$ and $BP = 3$

Find the diameter of the circle.

- (b) In the figure, AOB is the diameter of a circle centre O . D is a point on chord AC such that $DA = DO$ and OD is produced to E . AF is the bisector of $\angle BAC$ and cuts BE in G .



Prove that:

(i) $GA = GB$

3

(ii) $AOGE$ is a cyclic quadrilateral

2

- (c) (i) Prove that $\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$

2

(ii) Hence find $\int 2\sin^3 \theta \, d\theta$

3

Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

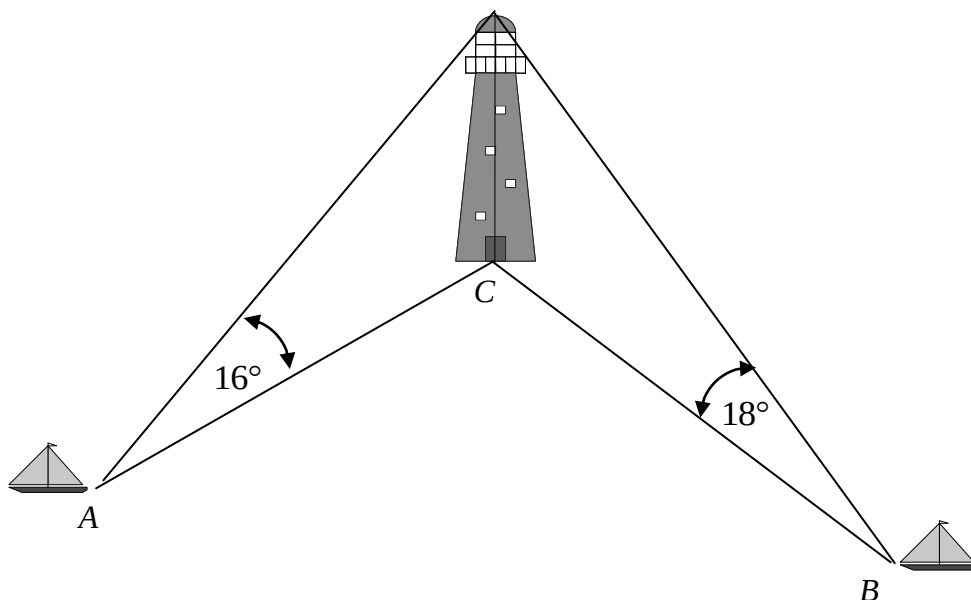
- (a) A boat sails from a point A to a point B .

At point A the captain of the ship measures the angle of elevation of the top of a lighthouse as 16° and the bearing of the lighthouse as 040° .

At point B the captain of the ship measures the angle of elevation of the top of the lighthouse as 18° and the bearing of the lighthouse as 340° .

The top of the lighthouse is known to be 80 m above sea level.

The diagram below shows the angles of elevation of the top of the lighthouse from A and B .

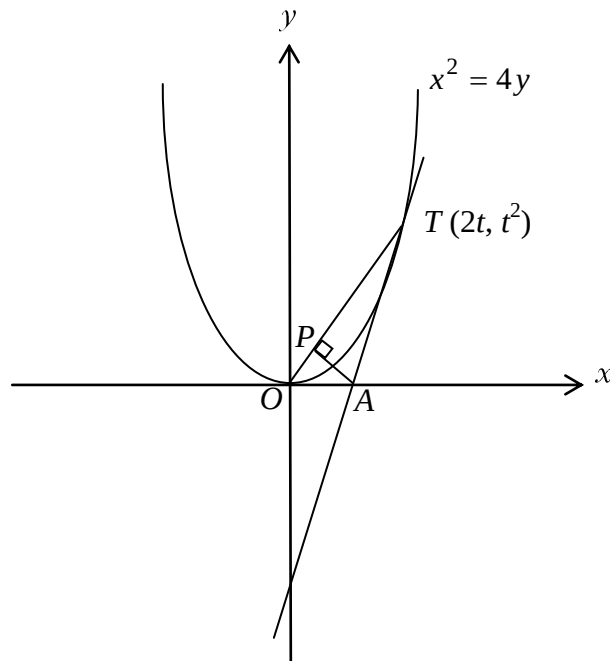


- | | | |
|-------|--|----------|
| (i) | Draw a bearing diagram showing the relative positions of A , B and C and use your diagram to explain why $\angle ACB = 60^\circ$. | 1 |
| (ii) | Hence, find the distance between A and B , correct to the nearest metre. | 3 |
| (iii) | Hence, find the bearing of B from A , to the nearest degree. | 2 |

Question 5 continued on page 7

Question 5 continued

Marks



- (b) The tangent at $T(2t, t^2)$, $t \neq 0$, on the parabola $x^2 = 4y$ meets the x axis at A .

$P(x, y)$ is the foot of the perpendicular from A to OT , where O is the origin.

The equation of the tangent at T is $y = tx - t^2$

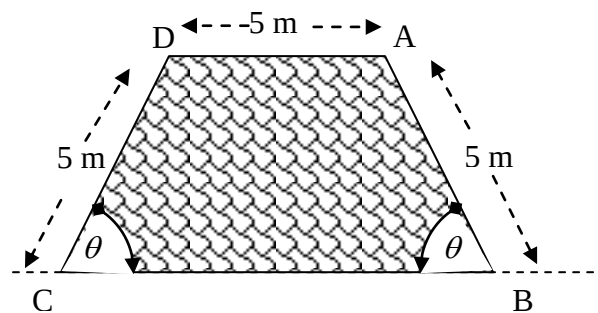
- | | |
|--|----------|
| (i) Prove that the equation of AP is $y = -\frac{2}{t}(x - t)$ | 2 |
| (ii) Show that the equation of OT is $t = \frac{2y}{x}$ | 1 |
| (iii) Hence, or otherwise, prove that the locus of $P(x, y)$ lies on a circle with centre $(0,1)$ and give its radius. | 3 |

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet **Marks**

- (a) (i) Draw a neat sketch of the function, $y = \log_e(x-1)$. 1
- (ii) This function meets the line $y = 2$ at the point P and the x -axis at the point Q . Show that the coordinates of P and Q are $(e^2 + 1, 2)$ and $(2, 0)$ respectively. 2
- (iii) If S is the point $(0, 2)$, find the co-ordinates of the point R if $OSPR$ is a rectangle. Label the points S and R on your sketch. 1
- (iv) Show that the arc PQ , divides the rectangle $OSPR$ into two regions of equal area. 3

- (b) The illustration below is part of the cross section of the roof of the Mathematics Faculty staffroom.



- (i) If $\angle ABC = \angle DCB = \theta$ show that the area of this cross section is given by: 2

$$A = 25 \sin \theta (1 + \cos \theta)$$

- (ii) Given $\frac{dA}{d\theta} = 50 \cos^2 \theta + 25 \cos \theta - 25$ 3
and $\frac{d^2A}{d\theta^2} = -100 \cos \theta \sin \theta - 25 \sin \theta$

Find the value of θ which will make this area a maximum.

End of Examination



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SOLUTIONS

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Question	Answer	Marks
1(a)	$\frac{3}{2-3x} \leq \frac{2}{3}$ $2-3x \neq 0$ $\therefore x = \frac{2}{3} \quad \boxed{\checkmark}$ $\frac{3}{2-3x} = \frac{2}{3}$ $9 = 4 - 6x$ $-6x = 5$ $x = \frac{-5}{6} \quad \boxed{\checkmark}$ Test points $\quad \boxed{\checkmark} \quad \frac{-5}{6} \quad \boxed{\times} \quad \frac{2}{3} \quad \boxed{\checkmark}$ $\therefore x \leq \frac{-5}{6} \quad \text{and} \quad x > \frac{2}{3} \quad \boxed{\checkmark}$	3
1(b)	$\sin 2\theta = \cos \theta$ $2 \sin \theta \cos \theta = \cos \theta$ $2 \sin \theta \cos \theta - \cos \theta = 0$ $\cos \theta (2 \sin \theta - 1) = 0 \quad \boxed{\checkmark}$ $\cos \theta = 0 \quad \text{or} \quad 2 \sin \theta - 1 = 0$ $\theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \quad \boxed{\checkmark} \quad \sin \theta = \frac{1}{2}$ $\theta = \frac{\pi}{6} \text{ or } \frac{5\pi}{6} \quad \boxed{\checkmark}$	3
1(c)	$\lim_{\theta \rightarrow 0} \frac{\tan \frac{\theta}{3}}{\theta} = \frac{\tan \frac{\theta}{3}}{\frac{\theta}{3}} \times \frac{\frac{\theta}{3}}{\theta} \quad \boxed{\checkmark}$ $= 1 \times \frac{1}{3}$ $= \frac{1}{3} \quad \boxed{\checkmark}$	2

1(d)	$\frac{dy}{dx} = \frac{f(x+h) - f(x)}{h}$ $= \frac{2(x+h)^2 + (x+h) - (2x^2 + x)}{h}$ $= \frac{4xh + h^2 + h}{h}$ $= \frac{h(4x^2 + h + 1)}{h}$ ☒ $\lim_{x \rightarrow 0} = 4x^2 + 0 + 1$ $= 4x^2 + 1$ ☒	2
1(e)	(1, 4) and (3, -6) externally 4:1 $\therefore \frac{1 \times 1 + -4 \times 3}{1 - 4}, \frac{1 \times 4 + -4 \times -6}{1 - 4}$ ☒ $\therefore P \left(\frac{11}{3}, \frac{-28}{3} \right)$ ☒	2

Question	Answer	Marks
2(a)	$= \int_1^{\sqrt{3}} 6x\sqrt{1+x^2} dx \quad u = 1+x^2$ $= \int 6x\sqrt{u} \frac{du}{2x} \quad \frac{du}{dx} = 2x$ $= \int 3\sqrt{u} du \quad \therefore dx = \frac{du}{2x}$ $= \int 3u^{\frac{1}{2}} du \quad \checkmark$ $= \left[\frac{3u^{\frac{3}{2}}}{\frac{3}{2}} \right]$ $= \left[2(1+x^2)^{\frac{3}{2}} \right]_1^{\sqrt{3}} \quad \checkmark$ $= \left[2(1+(\sqrt{3})^2)^{\frac{3}{2}} \right] - \left[2(1+1^2)^{\frac{3}{2}} \right]$ $= 16 - 4\sqrt{2} \quad \checkmark$	3
2(b)	$f(x) = \ln(\tan x)$ $f'(x) = \frac{\sec^2 x}{\tan x} \quad \checkmark$ $= \frac{1}{\cos^2 x} \times \frac{\cos x}{\sin x}$ $= \frac{1}{\cos x \sin x}$ $= \frac{1}{\frac{1}{2} \sin 2x} \quad \checkmark$ $= 2 \operatorname{cosec} 2x \quad \checkmark$	3
2(c)(i)	$\cos 3t - \sqrt{3} \sin 3t$ $\therefore R = \sqrt{1^2 + \sqrt{3}^2} = 2$ $\tan \alpha = \frac{\sqrt{3}}{1} \quad \text{hence} \quad \alpha = \frac{\pi}{3} \quad \checkmark$ $\therefore \cos 3t - \sqrt{3} \sin 3t = 2 \cos \left(3t + \frac{\pi}{3} \right) \quad \checkmark$	2

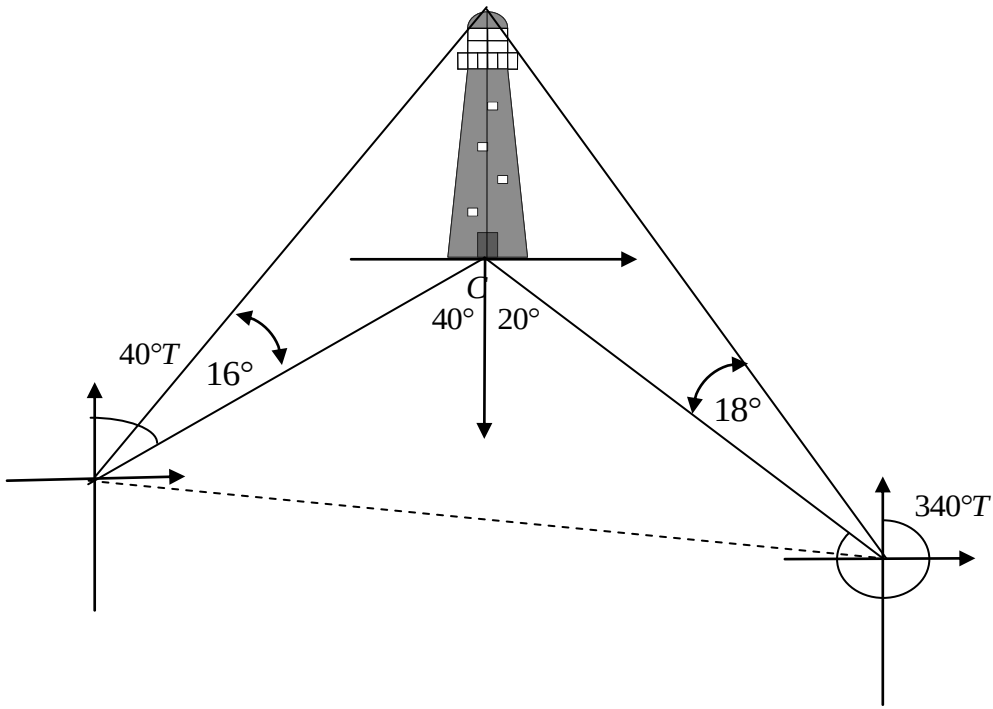
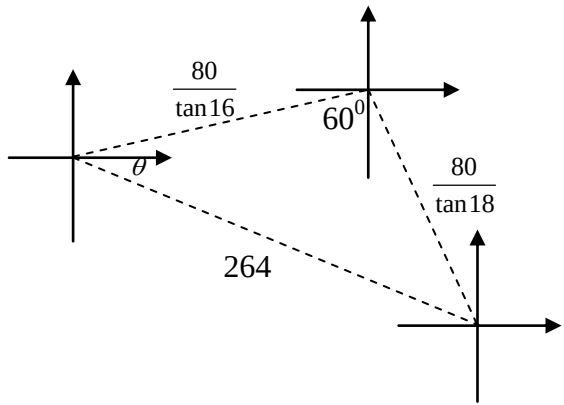
2(c)(ii)	$\cos 3t - \sqrt{3} \sin 3t = 2 \cos \left(3t + \frac{\pi}{3} \right)$ $\therefore \text{amplitude} = 2$ ☒ $\text{period} = \frac{2\pi}{3}$ ☒	2
2(d)	<i>Equations = 9 letters (5 vowels, 4 consonants)</i> <i>$\therefore 7 \text{ letters} = 5 \text{ vowels and } 2 \text{ consonants}$</i> <i>$\therefore {}^5P_5 \times {}^4P_2$</i> ☒ <i>$= 1440 \text{ ways}$</i> ☒	2

Question	Answer	Marks
3(a)	$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$ step1: prove true for $n = 2$ $LHS = 1 - \frac{1}{2^2} = \frac{3}{4} \qquad RHS = \frac{2+1}{2 \times 2} = \frac{3}{4}$ $\therefore LHS = RHS$ (hence true for $n = 2$) ☑ Step 2: assume true for $n = k$ $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}$ Step 3: prove true for $n = k+1$ $RTP: \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2(k+1)}$ $ \begin{aligned} LHS &= \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) \\ &= \frac{k+1}{2k} \left(1 - \frac{1}{(k+1)^2}\right) \\ &= \frac{k+1}{2k} \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right) \\ &= \frac{k+1}{2k} \left(\frac{k(k+2)}{(k+1)^2}\right) \\ &= \frac{1}{2} \left(\frac{k+2}{(k+1)}\right) \\ &= \frac{k+2}{2(k+1)} \\ &= RHS \end{aligned} $ $\therefore$ true for $n = k+1$ ☑ Step 4: since true for $n=2$ and $n=2+1=3, \dots$ and true for $n = k$ and $n = k+1$ therefore it is true for $n \geq 2$ ☑	3

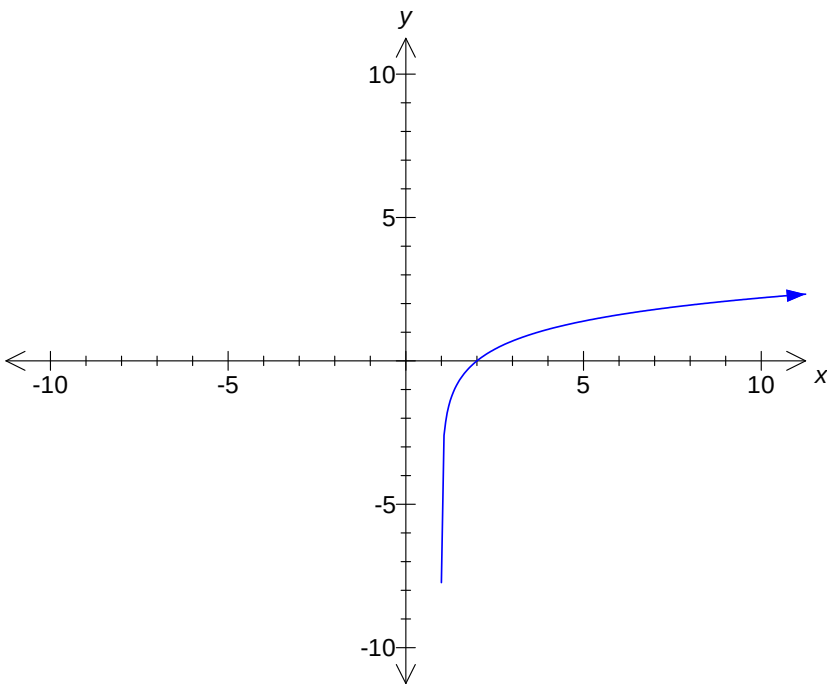
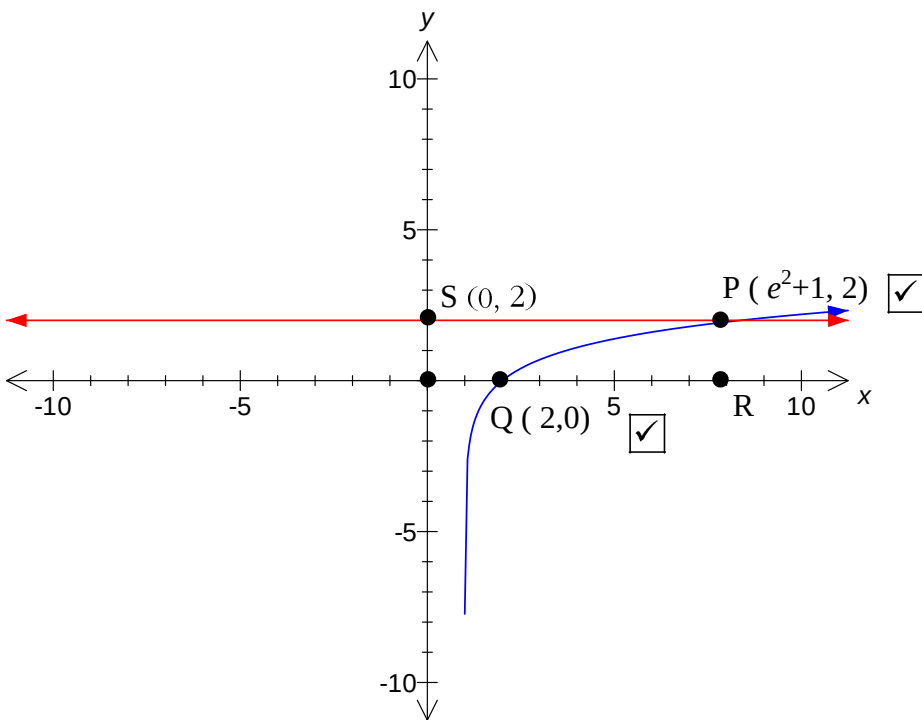
3(b)	$\int \sin^2 3x \, dx = \frac{1}{2}x - \frac{1}{12}\sin 6x \quad \checkmark$ $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 3x \, dx = \left[\frac{1}{2}x - \frac{1}{12}\sin 6x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$ $= \left[\frac{1}{2}\left(\frac{\pi}{3}\right) - \frac{1}{12}\sin 6\left(\frac{\pi}{3}\right) \right] - \left[\frac{1}{2}\left(\frac{\pi}{4}\right) - \frac{1}{12}\sin 6\left(\frac{\pi}{4}\right) \right] \quad \checkmark$ $= \left(\frac{\pi}{6} - 0 - \frac{\pi}{8} - \frac{1}{12} \right)$ $= \frac{\pi}{24} - \frac{1}{12} \quad \checkmark$	3
3(c)	$p(x) = x^3 + ax^2 + bx + 12$ $p(-1) = -1 + a - b + 12 = 0 \quad \therefore a - b = -11 \quad (i)$ $p(-2) = -8 + 4a - 2b + 12 = 8 \quad \therefore 4a - 2b = 4 \quad (ii) \quad \checkmark$ $(ii) - 2(i): \quad 2a = 26$ $\boxed{a = 13} \quad \checkmark$ $\text{sub } a = 13 \text{ into } (i)$ $\therefore 13 - b = -11$ $-b = -24$ $\boxed{b = 24} \quad \checkmark$	3
3(d)	$x^3 - 6x^2 - 2x + 4 = 0$ $(i) \quad \alpha + \beta + \gamma = \frac{-b}{a} = 6 \quad \checkmark$ $(ii) \quad \alpha^2 + \beta^2 + \gamma^2$ $= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \quad \checkmark$ $= (6)^2 - 2(-2)$ $= 36 + 4$ $= 40 \quad \checkmark$	3

Question	Answer	Marks
4(a)	Using similar triangles: $\frac{4}{d+3} = \frac{3}{9} \quad (\text{where } d = \text{diameter}) \quad \checkmark$ $3d + 9 = 36$ $3d = 27$ $d = 9$ $\therefore \text{diameter} = 9 \quad \checkmark$	3
4(b)(i)	$OB = OE$ (equal radii) $\angle OBE = \angle OEB$ (base $\angle$'s of isocles Δ are $=$) $\checkmark$ $\angle OEB = \angle BAC$ (angles standing on same arc are equal) $\checkmark$ $\therefore \angle OBE = \angle BAC$ $\therefore \Delta AGB$ is isocles (base $\angle$'s of isocles Δ are $=$) $\checkmark$ $\therefore AG = GB$	3
4(b)(ii)	$\angle OAG = \angle GEO$ (angles standing on same arc are equal) label X (intersection of AF and OE) $\therefore \angle AXO = \angle EXG$ (vertically opp $\angle$'s $=$) $\checkmark$ $\therefore \angle AOX = \angle EGX$ (angle sum of a Δ is supplementary) since $\angle AOX = \angle EGX$ $\therefore$ angles standing on the same arc are equal $\checkmark$ $\therefore AOGE$ is a cyclic quadrilateral	2
4(c)(i)	$\sin 3\theta = \sin(2\theta + \theta)$ $= \sin 2\theta \cos \theta + \sin \theta \cos 2\theta \quad \checkmark$ $= (2 \sin \theta \cos \theta)(\cos \theta) + \sin \theta(1 - 2 \sin^2 \theta)$ $= 2 \sin \theta \cos^2 \theta + \sin \theta - 2 \sin^3 \theta$ $= 2 \sin \theta(1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta \quad \checkmark$ $= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta$ $= 3 \sin \theta - 4 \sin^3 \theta$	2

4(c)(ii)	$\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$ $4\sin^3 \theta = 3\sin \theta - \sin 3\theta$ $\sin^3 \theta = \frac{1}{4}(3\sin \theta - \sin 3\theta) \quad \boxed{\checkmark}$ $\int 2\sin^3 \theta \, d\theta = 2 \int \sin^3 \theta \, d\theta$ $= 2 \int \frac{1}{4}(3\sin \theta - \sin 3\theta) \, d\theta \quad \boxed{\checkmark}$ $= \frac{1}{2} \int (3\sin \theta - \sin 3\theta) \, d\theta$ $= \frac{1}{2} \left[-3\cos \theta + \frac{1}{3}\cos 3\theta \right]$ $= -\frac{3}{2}\cos \theta + \frac{1}{6}\cos 3\theta + C \quad \boxed{\checkmark}$	3
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Question	Answer	Marks
5(a)(i)	 ☒ alternate $\angle$'s = $\therefore \angle ACB = 20 + 40 = 60^\circ$	1
5(a)(ii)	Let $AC = x$, $BC = y$ and $CT = 80$ m $\therefore \tan 16 = \frac{80}{x}$, $\tan 18 = \frac{80}{y}$ and $AB^2 = x^2 + y^2 - 2xy \cos 60$ ☒ $AB^2 = \left(\frac{80}{\tan 16} \right)^2 + \left(\frac{80}{\tan 18} \right)^2 - 2 \left(\frac{80}{\tan 16} \right) \left(\frac{80}{\tan 18} \right) \cos 60$ $AB^2 = 80^2 \left(\frac{1}{\tan^2 16} + \frac{1}{\tan^2 18} - \frac{2 \cos 60}{\tan 16 \tan 18} \right)$ $AB^2 = 69766.6396$ $AB = \sqrt{69766.6396}$ $AB = 264 \text{ m (nearest metre)}$ ☒ ☒	3
5(a)(iii)	 $\frac{\sin \theta}{\left(\frac{80}{\tan 18} \right)} = \frac{\sin 60}{264}$ $\therefore \theta = \sin^{-1} \left(\frac{\sin 60}{264} \times \frac{80}{\tan 18} \right)$ $\theta = 54^\circ \text{ (nearest degree)}$ ☒ $\therefore \text{Bearing of B from A}$ $= 54 + 40 = 090^\circ T$ ☒	2

5(b)(i)	$m_{OT} = \frac{t^2 - 0}{2t - 0} = \frac{t}{2}$ $AP \perp OT$ $\therefore m_{AP} = -\frac{2}{t} \quad \boxed{\checkmark}$ $A \text{ is where tangent } y = tx - t^2 \text{ crosses } x \text{ axis} \quad \therefore A(t, 0) \quad \boxed{\checkmark}$ $\therefore \text{equation } AP : y - 0 = -\frac{2}{t}(x - t)$ $y = -\frac{2}{t}(x - t)$	2
5(b)(ii)	$m_{OT} = \frac{t^2 - 0}{2t - 0} = \frac{t}{2}$ $\therefore \text{Equation } OT \text{ is: } y - t^2 = \frac{t}{2}(x - 2t) \quad \boxed{\checkmark}$ $y - t^2 = \frac{tx}{2} - t^2$ $y = \frac{tx}{2}$ $\therefore tx = 2y$ $t = \frac{2y}{x}$	1
5(b)(iii)	$P \text{ lies on the line } AP : y = -\frac{2}{t}(x - t)$ $\text{and } t = \frac{2y}{x}$ $\therefore y = \frac{-2}{\frac{2y}{x}} \left(x - \frac{2y}{x} \right) \quad \boxed{\checkmark}$ $y = \frac{-2x}{2y} \left(x - \frac{2y}{x} \right)$ $y = \frac{-x}{y} \left(x - \frac{2y}{x} \right)$ $y^2 = -x^2 + 2y$ $x^2 - 2y + y^2 = 0$ $(x - 1)^2 + y^2 = 1$ $\therefore \text{circle with centre } (1, 0) \quad \boxed{\checkmark}$ $\therefore \text{radius} = 1 \quad \boxed{\checkmark}$	3

Question	Answer	Marks
6(a)(i)		1
6(a)(ii)		2
6(a)(iii)	$R (e^2 + 1, 0)$ ✓	1

6(a)(iv)	$Area \text{ rectangle} = L \times B = 2(e^2 + 1)$ $y = \log_e(x-1) \quad \therefore x = e^y + 1$ $\int_0^2 e^y + 1 \, dy = [e^y + y]_0^2 \quad \checkmark$ $= [(e^2 + 2) - (e^0 + 0)]$ $= [(e^2 + 2) - (e^0 + 0)]$ $= [(e^2 + 2) - 1]$ $= e^2 + 1 \quad \checkmark$ $\therefore Area \text{ arc } PQ = e^2 + 1$ $Area \text{ arc} = \frac{1}{2}(\text{area rectangle}) \quad \checkmark$	3
6(b)(i)	Area of trapezium = area rectangle + 2 area of triangles $A = 5y + 2 \times \frac{1}{2}xy \quad \text{where } \sin \theta = \frac{y}{5} \Rightarrow y = 5 \sin \theta \quad \checkmark$ $\cos \theta = \frac{x}{5} \Rightarrow x = 5 \cos \theta$ $\therefore A = 25 \sin \theta + (5 \cos \theta)(5 \sin \theta)$ $A = 25 \sin \theta (1 + \cos \theta) \quad \checkmark$	2
6(b)(ii)	Max occurs when $\frac{dA}{d\theta} = 0$ $\therefore \frac{dA}{d\theta} = 50 \cos^2 \theta + 25 \cos \theta - 25 = 0$ $25(2 \cos^2 \theta + \cos \theta - 1) = 0$ $25(2 \cos \theta - 1)(\cos \theta + 1) = 0$ $\therefore 2 \cos \theta - 1 = 0 \quad \text{or} \quad \cos \theta + 1 = 0$ $2 \cos \theta = 1 \quad \cos \theta = -1$ $\theta = 60 \text{ or } 300 \quad \theta = 270 \quad \checkmark$ since θ is acute $\therefore \text{test } \theta = \frac{\pi}{3} \quad \checkmark$ $\frac{d^2 A}{d\theta^2} = -100 \cos 60 \sin 60 - 25 \sin 60 < 0 \quad \therefore \text{concave up since } \frac{d^2 A}{d\theta^2} < 0$ $\therefore \text{max of } A \text{ at } \theta = \frac{\pi}{3} \quad \checkmark$	3